ADAPTIVE MULTI-JUNCTION TRAFFIC GRID CONTROLLER

Task 5 – Full Verification Report

ModelSim simulation results | 07 November 2026

1. Objective

Task 5 aimed to verify the integrated AMTGC design using a unified self-checking Verilog testbench, tb_amtgc_system. The reported tests covered reset, normal traffic, adaptive density inputs, pedestrian arbitration, emergency handling, and randomized traffic conditions.

2. Verification Environment

Verification was performed in Intel/Altera ModelSim. The project folder was C:/AMTGC_Task5_Verification/. The reported top-level testbench was tb_amtgc_system.v, with the design modules amtgc_top.v, junction_controller.v, generic_timer.v, and ped_arbiter.v.

File	Purpose
amtgc_top.v	Integrates the two junction controllers and pedestrian arbiter.
junction_controller.v	Controls junction states, traffic lights, pedestrian phases, and adaptive green-time logic.
generic_timer.v	Provides parameterized timing for traffic and pedestrian phases.
ped_arbiter.v	Arbitrates pedestrian requests for Junction A and Junction B.
tb_amtgc_system.v	Runs integrated self-checking verification tests.

3. Tests Performed

Test	Description	Reported result
1. Initial reset	Reset applied at simulation start.	Completed
2. Normal traffic	Normal traffic operation exercised.	Completed
3. Adaptive density	Calculated green times checked at density levels 0, 1, 3, 5, and 7.	PASS
4. Pedestrian fairness	120 completed services counted while both requests were active.	PASS
5. Emergency override	Emergency override test sequence executed.	Completed
6. Repeated emergency	Repeated emergency assertion/deassertion sequence executed.	Completed
7. Active reset	Reset asserted during active simulation.	Completed
8. Randomized traffic	500 randomized test cycles executed.	Completed

4. Adaptive Timing Results

Traffic density	Calculated green time (cycles)
0	10
1	13
3	19
5	25
7	30 (maximum clamp)

The testbench reported expected calculated values of 10, 13, 19, 25, and 30 cycles for the selected density values. This confirms the calculation values reported by the testbench; it does not independently prove that measured green-phase durations in the waveform matched those values.

5. Pedestrian Arbitration and Fairness

The testbench reported 120 completed pedestrian services: 60 for Junction A and 60 for Junction B. The service-count difference was zero, satisfying the fairness condition programmed into this testbench run.

6. Simulation Summary

Metric	Reported value
Total safety checks	2,820
Total errors	0
Pedestrian services	120 (A: 60, B: 60)
Randomized cycles	500
Simulation finish time	28,190 ns
Final testbench message	AMTGC TASK 5: PASS

The supplied ModelSim transcript reports 2,820 safety checks, zero errors, and the final message AMTGC TASK 5: PASS. The simulation finished at 28,190 ns.

7. Coverage Notes and Limitations

The transcript confirms that the listed sequences ran and that the testbench's programmed checks reported no errors. However, the supplied summary does not show a dedicated assertion proving green-wave coordination across at least five density combinations, nor does it prove that actual measured green-phase durations matched the calculated values. These items should be explicitly checked if they are mandatory acceptance criteria.

Evidence to retain: the complete ModelSim Transcript with the final summary; waveform screenshots for normal traffic, emergency all-red, pedestrian service/fairness, and high-density timing; and the final unified testbench source. Attach only screenshots actually captured from the simulation.

8. Conclusion

According to the supplied self-checking testbench output, this verification run completed with 2,820 safety checks, zero reported errors, 120 pedestrian services split equally between the two junctions, and 500 randomized cycles. The testbench printed AMTGC TASK 5: PASS. Green-wave coordination and measured adaptive phase durations should be verified separately before claiming that every assignment requirement has been demonstrated.

Prepared from the ModelSim results supplied for the Task 5 verification run.